# Telecommunication Systems
# ICE 3109

## Lecture 6
## Traffic Analysis

# Traffic Analysis

- Traffic in a communication network refers to the aggregate of all user requests being serviced by the network.
- The basic goal of traffic analysis is to provide a method for determining the cost effectiveness of various sizes and configuration of networks.
- The service requests arrive randomly and usually require unpredictable service times.

# Traffic Analysis

The techniques of traffic analysis can be divided into two general categories: loss systems and delay systems. The appropriate analysis category for a particular system depends on the system's treatment of overload traffic. In a loss system overload traffic is rejected without being serviced. In a delay system overload traffic is held in a queue until the facilities become available to service it. Conventional circuit switching operates as a loss system since excess traffic is blocked and not serviced without a retry on the part of the user. In some instances "lost" calls actually represent a loss of revenue to the carriers by virtue of their not being completed.

# Traffic Analysis

Store-and-forward message or packet switching obviously possesses the basic characteristics of a delay system. Sometimes, however, a packet-switching operation can also contain certain aspects of a loss system. Limited queue sizes and virtual circuits both imply loss operations during traffic overloads. Circuit-switching networks also incorporate certain operations of a delay nature in addition to the loss operation of the circuits themselves. For example, access to a digit receiver, an operator, or a call processor is normally controlled by a queuing process.

The basic measure of performance for a loss system is the probability of rejection (blocking probability). A delay system, on the other hand, is measured in terms of service delays. Sometimes the average delay is desired, while at other times the probability of the delay exceeding some specified value is of more interest.

# Traffic Characterization

**Traffic – Random in nature.**

**Following analysis required :**

- **Probability Theory**
- **Stochastic Process**

In the realm of applied mathematics, where these subjects are treated more formally, blocking probability analyses are referred to as congestion theory and delay analyses are referred to as queuing theory. These topics are also commonly referred to as traffic flow analysis. In a circuit-switched network, the "flow" of messages is not so much of a concern as are the holding times of common equipment. A circuit-switched network establishes an end-to-end circuit involving various network facilities (transmission links and switching stages) that are held for the duration of a call. From a network point of view, it is the holding of these resources that is important, not the flow of information within individual circuits.

# Traffic Measurement

- Traffic measurement in telecommunications is essential for assessing and managing the performance, capacity, and efficiency of communication networks.
- By measuring traffic, network operators can monitor demand, optimize resource usage, and ensure quality of service.
- Here are some key aspects of traffic measurement in telecommunications:
  - Traffic Metrics
  - Data Collection Techniques
  - Types of Traffic
  - Applications of Traffic Measurement
  - **Challenges in Traffic Measurement**

# Traffic Metrics

•**Erlang**: A basic unit of telecom traffic measurement. One Erlang represents one hour of continuous voice or data transmission. This metric helps to calculate the call-carrying capacity of a network.

1 Erlang= 36 CCS

Where, CCS= Century (hundred) Call Seconds

•**Busy Hour Traffic (BHT)**: The peak traffic volume during the busiest hour of a day. BHT is crucial for network dimensioning and planning to avoid congestion.

•**Call Attempt Rate (CAR)**: Measures the number of calls initiated in a network, giving insights into user demand.

•**Blocking Probability**: The likelihood that a call or data session will be blocked due to insufficient resources. This metric helps assess service quality, especially during peak times.

# Data Collection Techniques

- **Probes and Counters**: Installed at various points within the network to capture data traffic, such as call duration, call success rate, and data throughput.

- **Network Monitoring Tools**: Software solutions that monitor, collect, and analyze traffic metrics in real-time. Examples include Simple Network Management Protocol (SNMP) and NetFlow for IP-based traffic.

- **Billing Data**: Data collected from billing systems can also serve as a secondary source of traffic information, particularly for postpaid customers, where every call or data session generates billing records.

# Types of Traffic

- **Voice Traffic**: Measured in terms of call minutes and Erlangs. Voice traffic typically uses circuits and is managed differently from packet-based data.

- **Data Traffic**: Measured in bytes or packets, often requiring more complex analysis due to the variability in traffic patterns and the need for higher data rates.

# Applications of Traffic Measurement

- **Capacity Planning**: Helps in predicting future demand and expanding network capacity accordingly.

- **Quality of Service (QoS)**: Ensures consistent service quality by monitoring metrics like latency, jitter, and packet loss.

- **Network Optimization**: Identifies underutilized or congested parts of the network to optimize routing and resource allocation.

- **Fraud Detection**: Unusual traffic patterns can indicate fraudulent activities, prompting further investigation.

# Challenges in Traffic Measurement

- **Scalability**: Handling large volumes of data traffic across global networks.

- **Dynamic Traffic Patterns**: The rise in video streaming, IoT, and other services makes traffic patterns highly variable.

- **Privacy Concerns**: Collecting detailed traffic data raises privacy issues, requiring compliance with regulations like GDPR.

# Traffic Measurement

Two important parameters used to characterize traffic are the average arrival rate $\lambda$ and the average holding time $t_m$. If the traffic intensity $\lambda$ is expressed in erlangs, then

$$A = \lambda t_m$$

where $\lambda$ and $t_m$ are expressed in like units of time (e.g., calls per second and seconds per call, respectively).

# Traffic Measurement

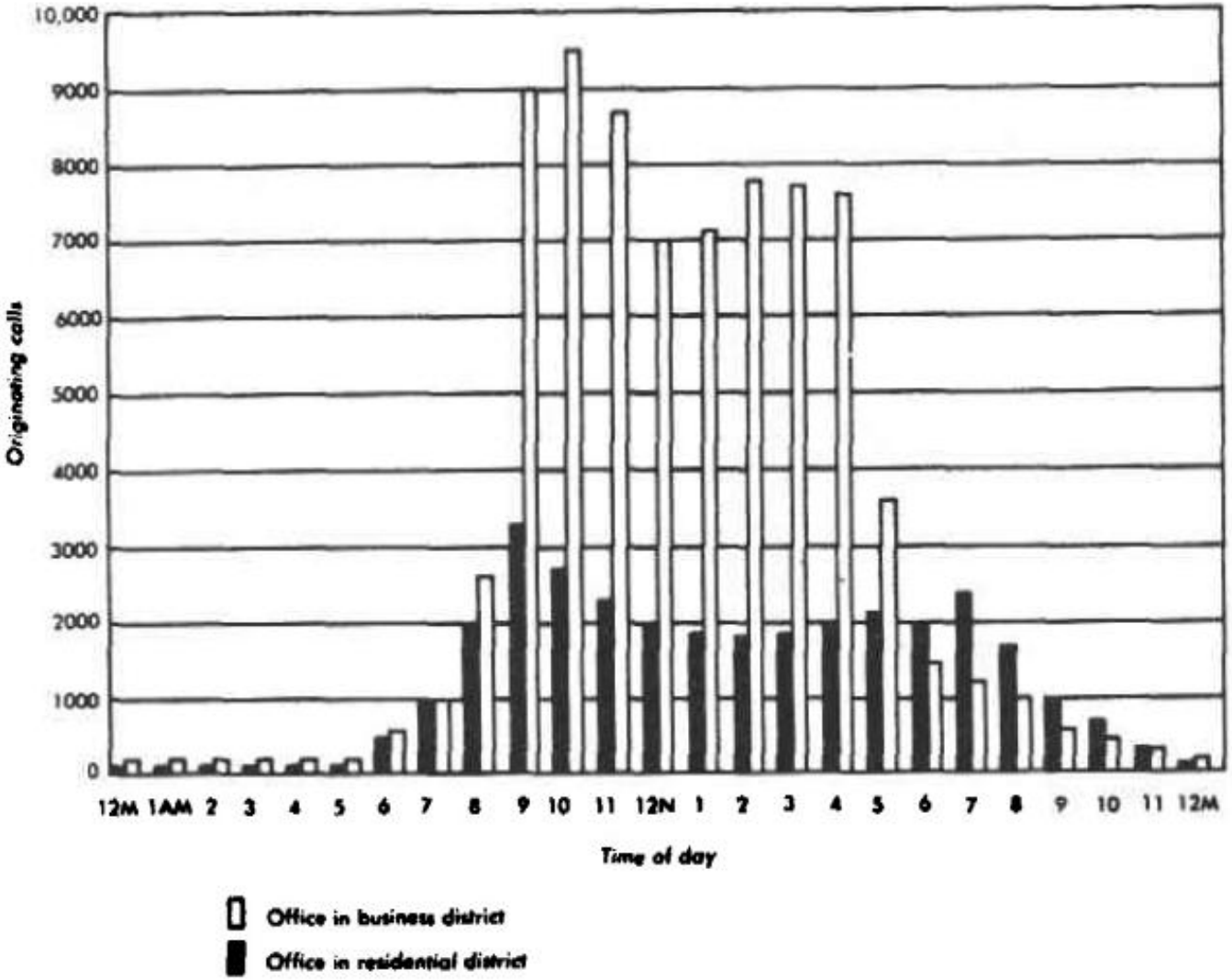



**Figure 12.2** Traffic volume dependence on time of day.

# Arrival Distributions

The most fundamental assumption of classical traffic analysis is that call arrivals are independent. That is, an arrival from one source is unrelated to an arrival from any other source. Even though this assumption may be invalid in some instances, it has general usefulness for most applications. In those cases where call arrivals tend to be correlated, useful results can still be obtained by modifying a random arrival analysis. In this manner the random arrival assumption provides a mathematical formulation that can be adjusted to produce approximate solutions to problems that are otherwise mathematically intractable.

# Arrival Distributions

***Negative Exponential Interarrival Times***

Designate the average call arrival rate from a large group of independent sources (subscriber lines) as λ. Use the following assumptions:

1. Only one arrival can occur in any sufficiently small interval.
2. The probability of an arrival in any sufficiently small interval is directly proportional to the length of the interval. (The probability of an arrival is $\lambda \Delta t$, where $\Delta t$ is the interval length.)
3. The probability of an arrival in any particular interval is independent of what has occurred in other intervals.

**Probability distribution of interval time is**

$$P_0(\lambda t) = e^{-\lambda t}$$

It defines the probability that no arrival occurs in a randomly selected interval t. This is identical to the probability that $t$ seconds elapse from one arrival to the next

# Arrival Distributions

**Example 12.1.** Assuming each of 10,000 subscriber lines originate one call per hour, how often do two calls arrive with less than 0.01 sec between them?

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The average arrival rate is

$$\lambda = 3600/10{,}000 = 2.78 \text{ arrivals/sec}$$

the probability of no arrival in a 0.01-sec interval is

$$P_0(0.0278) = e^{-0.0278} = 0.973$$

Thus 2.7% of the arrivals occur within 0.01 sec of the previous arrival. Since the arrival rate is 2.78 arrivals per second, the rate of occurrence of interarrival times less than 0.01 sec is

$$2.78 \times 0.027 = 0.075 \text{ times/sec}$$

# Poisson Arrival Distribution

- The probability of j arrivals in an interval t can be determined as,

$$\boldsymbol{P_j(\lambda t) = \frac{(\lambda t)^j}{j!} e^{-\lambda t}}$$

- The equation is known as Poisson probability law.
- Arrivals are independent and occur at a given average rate of $\lambda$
- Poisson probability distribution should only be used for arrivals for large number of independent sources.

When $j = 0$ , the probability of no arrivals in an interval $t$ is $P_0(t)$. It can be obtained as

$$\boldsymbol{P_0(\lambda t) = e^{-\lambda t}}$$

# Poisson Arrival Distribution

$$P_j(\lambda t) = \frac{(\lambda t)^j}{j!} e^{-\lambda t}$$

- The equation represents the probability of experiencing exactly $j$ arrivals in $t$ seconds.
- There is more interest in determining the probability of $j$ or more arrivals in $t$ seconds.

$$P_{\geq j}(\lambda t) = \sum_{i=j}^{\infty} P_i(\lambda t)$$

$$= 1 - \sum_{i=0}^{j-1} P_i(\lambda t)$$

$$= 1 - P_{<j}(\lambda t)$$

# Poisson Arrival Distribution

**Example 12.2.** Given a message-switching node that normally experiences four arrivals per minute, what is the probability that eight or more arrivals occur in an arbitrarily chosen 30-sec interval?

# Poisson Arrival Distribution

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The average number of arrivals in a 30-sec interval is

$$\lambda t = 4 \times \frac{30}{60} = 2$$

The probability of eight or more arrivals (when the average is 2) is

$$P_{\geq 8}(2) = \sum_{i=8}^{\infty} P_i(2)$$

# Poisson Arrival Distribution

$$= 1 - \sum_{i=0}^{7} P_i(2)$$

$$= 1 - e^{-2}\left(1 + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \ldots + \frac{2^7}{7!}\right)$$

$$= 0.0011$$

# Poisson Arrival Distribution

**Example 12.3.** What is the probability that a 1000-bit data block experiences exactly four errors while being transmitted over a transmission link with a bit error rate (BER) of $10^{-5}$?

# Poisson Arrival Distribution

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***Solution.*** Assuming independent errors (a questionable assumption on many transmission links), we can obtain the probability of exactly four errors directly from the Poisson distribution. The average number of errors (arrivals) $\lambda t = 10^3 \times 10^{-5} = 0.01$. Thus

$$\text{prob(4errors)} = P_4(0.01) = \frac{(0.01)^4}{4!} e^{-0.01} = 4.125 \times 10^{-10}$$

# Holding Time Distributions

- Another important factor of traffic intensity is average holding time $t_m$.
- To determine blocking probabilities in a loss system / delay system, holding times need to know.
- Probability distribution of holding time is required to obtain the desired results.

Types:

1. Constant Holding Times
2. Exponential Holding Times

# Constant Holding Times

Although constant holding times cannot be assumed for conventional voice conversations, it is a reasonable assumption for such activities as per-call call processing requirements, interoffice address signaling, operator assistance, and recorded message playback. Furthermore, constant holding times are obviously valid for transmission times in fixed-length packet networks.

When constant holding time messages are in effect, it is straightforward to use Equation 12.3 to determine the probability distribution of active channels. Assume, for the time being, that all requests are serviced. Then the probability of $j$ channels being busy at any particular time is merely the probability that $j$ arrivals occurred in the time interval of length $t_m$ immediately preceding the instant in question. Since the average number of active circuits over all time is the traffic intensity $A = \lambda t_m$, the probability of $j$ circuits being busy is dependent only on the traffic intensity:

$$P_j(\lambda t_m) = P_j(A) = \frac{A^j}{j!} e^{-A}$$

# Exponential Holding Times

The most commonly assumed holding time distribution for conventional telephone conversations is the exponential holding time distribution:

$$P(>t) = e^{-\frac{t}{t_m}}$$

where $t_m$ is the average holding time. Equation 12.6 specifies the probability that a holding time exceeds the value $t$. This relationship can be derived from a few simple assumptions concerning the nature of the call termination process. Its basic justification, however, lies in the fact that observations of actual voice conversations exhibit a remarkably close correspondence to an exponential distribution.

The exponential distribution possesses the curious property that the probability of a termination is independent of how long a call has been in progress. That is, no matter how long a call has been in existence, the probability of it lasting another $t$ seconds is defined by Equation 12.6. In this sense exponential holding times represent the most random process possible. Not even knowledge of how long a call has been in progress provides any information as to when the call will terminate.

# Exponential Holding Times

Combining a Poisson arrival process with an exponential holding time process to obtain the probability distribution of active circuits is more complicated than it was for constant holding times because calls can last indefinitely. The final result, however, proves to be dependent on only the average holding time. Thus Equation 12.5 is valid for exponential holding times as well as for constant holding times (or any holding time distribution). Equation 12.5 is therefore repeated for emphasis: The probability of $j$ circuits being busy at any particular instant, assuming a Poisson arrival process and that all requests are serviced immediately, is

$$P_j(A) = \frac{A^j}{j!} e^{-A}$$

where $A$ is the traffic intensity in erlangs. This result is true for any distribution of holding times.